

I. DETAILED METHODOLOGY

This section gives the extended derivation of the proposed dual-branch Poincaré–Wasserstein Siamese framework that Section II of the main paper summarizes under space constraints. We proceed in the same order as the reference implementation: (A) problem formulation and episodic task construction, (B) pre-processing, (C) the shared Siamese encoder and its two heads, (D) the Poincaré ball and its differential-geometric operators, (E) hyperbolic prototype estimation, (F) geodesic prototypical classification, (G) the entropic Sinkhorn–Wasserstein structural loss on the local branch, (H) the composite training objective, and (I) the full episodic optimization procedure, given as pseudocode.

A. Problem Formulation and Episodic Task Construction

We address N -way K -shot classification of histopathology patches. A *task* (episode) $\mathcal{T}$ is drawn by sampling, for each of N classes $c \in \{1, \dots, N\}$, a support set $\mathcal{S}_c = \{x_{c,1}^s, \dots, x_{c,K}^s\}$ of K labeled images and a disjoint query set $\mathcal{Q}_c = \{x_{c,1}^q, \dots, x_{c,Q}^q\}$ of Q labeled images from the same class, with $\mathcal{S}_c \cap \mathcal{Q}_c = \emptyset$. The full support and query sets for the episode are $\mathcal{S} = \bigcup_{c=1}^N \mathcal{S}_c$ and $\mathcal{Q} = \bigcup_{c=1}^N \mathcal{Q}_c$.

To prevent optimistic leakage between model selection and final reporting, each class’s image pool is first split, *before any episodic sampling*, into a disjoint TRAIN-POOL and TEST-POOL (70/30 in our runs; $N_{\text{test-per-class}} = 30$ for BACH). During training, every episode draws its support *and* query images exclusively from the train-pool with light stochastic augmentation (horizontal/vertical flips, 90° rotations). During evaluation, episodes draw support from the train-pool (playing the role of the “ K labeled examples the clinician has on hand”) and queries exclusively from the held-out test-pool, so that no query image has ever participated in gradient updates, directly or through a prototype the model was trained to match. We additionally report a “full-pool” pass in which *every* image in a class’s test-pool is used as a query simultaneously (rather than $Q=5$ per episode) to compute the confusion matrices and t-SNE projections in Section III, giving a lower-variance estimate than any single sampled episode.

B. Pre-processing

Each histology tile is loaded, resized by area-interpolation to $224 \times 224 \times 3$, and contrast-enhanced with Contrast-Limited Adaptive Histogram Equalization (CLAHE) applied only to the lightness channel of the CIE-LAB representation (clip limit 2.0, tile grid 8×8), which equalizes local stain-intensity variation introduced by uneven H&E staining without amplifying color noise in the chrominance channels a^*, b^* . The image is then min-max normalized to $[0, 1]$ and standardized with the ImageNet channel statistics $\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$, matching the distribution the ResNet-18 backbone was pre-trained on. Support-set images additionally receive stochastic flip/rotation augmentation during training episodes only; query images and all test-time images are never augmented, so that accuracy is not inflated by test-time augmentation artifacts.

C. Shared Siamese Dual-Branch Encoder

A single weight-shared encoder f_θ is applied identically to every support and query image in an episode — the defining property of a Siamese network: there is no separate “query network,” so the embedding space in which distances are computed is guaranteed to be consistent across the handful of support examples and the unseen queries.

a) *Backbone.*: We use an ImageNet-pretrained ResNet-18, with its final global-average-pool and fully-connected classification layers removed, retaining only the convolutional stem. For an input $x \in \mathbb{R}^{3 \times 224 \times 224}$ this produces a feature tensor

$$F = \text{stem}_\theta(x) \in \mathbb{R}^{512 \times 7 \times 7}.$$

F is shared by both downstream heads, so the two branches differ only in how they read out from the same convolutional features — keeping the parameter budget close to a single ResNet-18 and making the local branch essentially free at inference time.

b) *Global head (hierarchy branch).*: F is spatially collapsed by global average pooling and passed through a single linear layer to a D -dimensional tangent vector,

$$v = W_g \text{GAP}(F) + b_g \in \mathbb{R}^D, \quad D=128,$$

which is then mapped onto the Poincaré ball by the exponential map at the origin (Eq. 1 below) to obtain the *global embedding* $z = \exp_0^c(v)$. Because z lies on a curved manifold whose volume grows exponentially with radius, a benign/malignant split that would need many Euclidean dimensions to keep siblings separated from cousins can be represented with low distortion in a comparatively low-dimensional z — this is the representation used for coarse, hierarchy-aware classification (Section I-D).

c) *Local head (structure branch).*: In parallel, F is passed through a 1×1 convolution reducing the channel dimension from 512 to $d=32$ without altering the 7×7 spatial grid, then adaptively average-pooled to a coarser 4×4 grid ($P=16$ points, $\text{grid} = \lfloor \sqrt{P} \rfloor$), and flattened into a set of P local tangent vectors $\{u_1, \dots, u_P\}$, $u_p \in \mathbb{R}^d$. Each u_p is independently mapped onto the same Poincaré ball, producing a set of P *local hyperbolic embeddings* $h_p = \exp_0^c(u_p)$ that retain coarse spatial layout information (each h_p corresponds to a $\sim 56 \times 56$ -pixel receptive-field neighborhood of the original patch) while being compact enough for optimal-transport comparison to remain cheap. Because the local head reads from the same F as the global head, the two branches are trained jointly and cannot diverge into inconsistent representations of the same input.

For Grad-CAM visualization (Section III of the main paper) we additionally expose F directly and back-propagate the norm $\|z\|$ of the global embedding through the last convolutional block, which highlights the tissue regions that most strongly push the embedding toward the boundary of the ball (i.e., toward a more “specific,” fine-grained class assignment).

D. The Poincaré Ball and its Operators

a) *The manifold.*: The Poincaré ball of curvature $-c$ ($c > 0$) in n dimensions is the open Euclidean ball equipped with a Riemannian metric that diverges at the boundary,

$$\mathbb{D}_c^n = \{x \in \mathbb{R}^n : c\|x\|^2 < 1\}, \quad g_x = \left(\frac{2}{1 - c\|x\|^2}\right)^2 g^{\text{Euclid}}.$$

As $\|x\| \rightarrow 1/\sqrt{c}$, the conformal factor $\lambda_x^c = 2/(1 - c\|x\|^2)$ blows up, so a fixed Euclidean step near the boundary corresponds to an ever-larger geodesic step: this is what lets a bounded disk hold an effectively exponentially growing volume with respect to geodesic radius, mirroring the branching factor of a hierarchical taxonomy (root $\rightarrow$ benign/malignant $\rightarrow$ subtype). All embedding norms are numerically clipped to stay at least $10^{-3}/\sqrt{c}$ inside the boundary (function `project_to_ball` in the implementation) to avoid the metric singularity in floating-point arithmetic.

b) *Exponential map at the origin.*: Since the encoder naturally outputs flat (Euclidean) vectors, we need a map from the tangent space at the origin, $T_0\mathbb{D}_c^n \cong \mathbb{R}^n$, onto the ball itself. The closed-form exponential map at the origin is

$$z = \exp_0^c(v) = \tanh(\sqrt{c}\|v\|) \frac{v}{\sqrt{c}\|v\|}, \quad (1)$$

implemented with an ε -guarded denominator ($\|v\| \mapsto \max(\|v\|, \varepsilon)$, $\varepsilon=10^{-6}$) for numerical safety when $v \rightarrow 0$. This map has the interpretable property that small-norm tangent vectors are sent close to the origin (broad, shared, “coarse” concepts near the root of the hierarchy) while large-norm tangent vectors are stretched toward the boundary (fine-grained, highly specific concepts), because $\tanh(\cdot)$ saturates while the metric’s conformal factor grows without bound at the same limit.

c) *Geodesic (Poincaré) distance.*: For two points x, y on the ball, the length of the shortest path between them under the metric above has the closed form

$$d_c(x, y) = \frac{1}{\sqrt{c}} \operatorname{arcosh} \left(1 + \frac{2c\|x - y\|^2}{(1 - c\|x\|^2)(1 - c\|y\|^2)} \right). \quad (2)$$

Because the denominator $(1 - c\|x\|^2)(1 - c\|y\|^2)$ shrinks toward zero as either point approaches the boundary, the same Euclidean gap $\|x - y\|$ produces a much larger geodesic distance for two boundary-adjacent, fine-grained-class points than for two origin-adjacent, coarse-class points — exactly the property required to keep visually similar but clinically distinct subtypes (e.g. lobular vs. ductal carcinoma) well separated once the network has learned to push them toward different regions of the boundary. In the implementation, the arcosh argument is clamped to $[1 + \varepsilon, \infty)$ and both denominator factors are clamped away from zero before the division, since Eq. 2 is otherwise numerically unstable close to the ball’s boundary. Pairwise distance matrices between a batch of N queries and M prototypes are computed by broadcasting Eq. 2 over all $N \times M$ pairs in a single vectorized call (`poincare_pairwise`).

d) *Logarithmic map at the origin.*: The inverse of Eq. 1, needed to bring ball points back into the (linear, averageable) tangent space, is

$$v = \log_0^c(z) = \frac{\operatorname{artanh}(\sqrt{c}\|z\|)}{\sqrt{c}\|z\|} z, \quad (3)$$

where artanh is evaluated as $\frac{1}{2}(\log(1+z) - \log(1-z))$ with its argument clamped to $(-1 + \varepsilon, 1 - \varepsilon)$ to avoid the log singularities at $z = \pm 1$.

E. Hyperbolic Prototype Estimation

Points on a curved manifold cannot be meaningfully averaged with a Euclidean arithmetic mean — the result would generally not even lie on the manifold, and would not minimize the natural (geodesic) notion of squared-distance dispersion. We instead compute an approximate *Karcher / Fréchet mean* of the K support embeddings of class c , $\{z_{c,1}^s, \dots, z_{c,K}^s\}$, via the standard log-average-exp (tangent-space, a.k.a. “Einstein midpoint”) construction:

$$p_c = \exp_0^c \left(\frac{1}{K} \sum_{k=1}^K \log_0^c(z_{c,k}^s) \right). \quad (4)$$

This is a one-step (as opposed to iterative Riemannian gradient descent) approximation to the true Karcher mean, which is exact for points close to the origin and remains an accurate, cheap, and *differentiable* surrogate for the small $K \in \{1, \dots, 10\}$ regime this work operates in, and critically avoids an inner optimization loop inside every training episode. The same construction, applied independently to each of the P spatial positions across the K support local embeddings, would give a local Karcher mean; however, for the local branch we instead retain the full *set* of local prototype points rather than collapsing them to a single mean per position, because the structural comparison in Section I-G needs a point cloud, not a centroid. In the implementation this “local prototype set” is formed as the Euclidean average of the K support local embeddings at each of the P grid positions, re-projected onto the ball to remain a valid (if slightly approximate) hyperbolic point set, $S_c = \{\bar{h}_{c,1}, \dots, \bar{h}_{c,P}\}$.

F. Geodesic Prototypical Classification

Given the N global class prototypes $\{p_1, \dots, p_N\}$ from Eq. 4 and a query global embedding z_q , the model computes the N geodesic distances $d_c(z_q, p_j)$ via Eq. 2 and classifies by a softmax over their *negatives* — i.e., closer prototypes receive higher probability, exactly as in Euclidean Prototypical Networks but with the flat metric replaced by the geodesic one:

$$P(y_q = c \mid x_q) = \frac{\exp(-d_c(z_q, p_c))}{\sum_{j=1}^N \exp(-d_c(z_q, p_j))}. \quad (5)$$

Because $p_1, \dots, p_N$ are recomputed from scratch, purely as a function of the current K support embeddings, at the start of *every* episode, no class-specific parameters are learned; only the shared encoder f_θ is optimized (Section I-H), which is what allows the resulting model to generalize, at test time, to

novel support sets — and in principle to classes not seen at all during training, provided the encoder’s induced geometry is faithful.

G. Local Structural Alignment via Entropic Sinkhorn–Wasserstein Distance

a) *Motivation.*: Two patches of the same subtype are not spatially registered: sectioning, mounting, and scanning do not fix a canonical orientation, so a duct that appears in grid-cell (1, 1) of one 4×4 local-embedding grid may appear in cell (3, 2) of another. Comparing the local branch’s $P=16$ features *index-to-index* (e.g. with a fixed sum of $d_c(h_p, g_p)$ over matched positions p) would then penalize this purely positional variance as if it were a class difference. Optimal transport instead finds the minimum-cost *matching* between the two P -point sets, which makes the comparison invariant to exactly this kind of shift/rotation/permutation of tissue structures across the grid.

b) *Entropy-regularized OT cost.*: Given the query’s local point set $\{h_p\}_{p=1}^P$ and a prototype local point set $\{g_q\}_{q=1}^P$ (from Eq. 4’s sibling construction), define the pairwise geodesic ground-cost matrix $C \in \mathbb{R}_{\geq 0}^{P \times P}$,

$$C_{pq} = d_c(h_p, g_q),$$

using Eq. 2 rather than Euclidean distance, so that the transport cost itself respects the hyperbolic geometry the local embeddings live on. Let μ, ν be the uniform distributions over the P query and P prototype points respectively (each local grid cell is given equal a-priori importance), and let $\Pi(\mu, \nu)$ be the set of transport plans $T \in \mathbb{R}_{\geq 0}^{P \times P}$ with row sums μ and column sums ν . The entropy-regularized OT cost is

$$\mathcal{W}_\varepsilon = \min_{T \in \Pi(\mu, \nu)} \langle T, C \rangle - \varepsilon H(T), \quad (6)$$

with $\langle T, C \rangle = \sum_{p,q} T_{pq} C_{pq}$ the total transport cost and $H(T) = -\sum_{p,q} T_{pq} \log T_{pq}$ the Shannon entropy of the plan. Subtracting $\varepsilon H(T)$ is what makes Eq. 6 tractable and, crucially, *differentiable* in C (and hence in the network parameters θ through h_p, g_q): without regularization the optimal T is a sparse, near-deterministic matching whose argmin structure is not smooth in C ; as ε grows, T spreads mass across more point pairs, trading matching sharpness for the smooth, matrix-multiplication-only solver below.

c) *Sinkhorn iterations.*: By Sinkhorn’s theorem, the entropically regularized optimal plan factorizes as $T^* = \text{diag}(u) K \text{diag}(v)$ for a kernel $K = \exp(-C/\varepsilon)$ (with a small additive floor 10^{-9} for numerical stability) and scaling vectors $u, v \in \mathbb{R}_{> 0}^P$ that are found by alternating row/column normalization,

$$u \leftarrow \frac{\mu}{K(\nu/K^\top u)}, \quad v \leftarrow \frac{\nu}{K^\top u}, \quad (7)$$

each division taken element-wise with a 10^{-9} floor in the denominator. We run 50 such iterations (fixed, unrolled, so the whole procedure is differentiable end-to-end via autograd — no implicit-function or custom-backward tricks are required), after which $\mathcal{W}_\varepsilon \approx \langle T^*, C \rangle$ is returned as the scalar structural

distance between the query’s local point set and a class’s local prototype set, denoted $\mathcal{W}_\varepsilon(h_q, S_c)$ for query image q (representing its full $\{h_p\}_{p=1}^P$) and class c ’s local prototype set S_c . We use $\varepsilon=0.1$ and $P=16$ throughout.

H. Composite Training Objective

Within an episode, the two branches are combined into a single scalar loss. The first term is the geodesic cross-entropy from Eq. 5, $\mathcal{L}_{CE} = -\sum_q \log P(y_q | x_q)$. The second term is a margin-based structural consistency loss over the local branch: for every query q with true class y_q , we draw one incorrect class $r \neq y_q$ uniformly at random and require the query’s local point set to be close (small $\mathcal{W}_\varepsilon$) to its own class’s local prototype set while being at least a margin m away from the wrong class’s local prototype set:

$$\mathcal{L}_W = \sum_q \left[\mathcal{W}_\varepsilon(h_q, S_{y_q}) + (m - \mathcal{W}_\varepsilon(h_q, S_r))_+ \right], \quad (8)$$

with $(\cdot)_+ = \max(\cdot, 0)$ the hinge and $m=1.0$. The total per-episode loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{CE} + \lambda \mathcal{L}_W, \quad \lambda=0.3, \quad (9)$$

matching Eq. 5 of the main paper. Only a *single* random negative class is sampled per query per step (rather than all $N-1$ negatives), which keeps the number of Sinkhorn solves per episode at $O(Q \cdot N)$ instead of $O(Q \cdot N^2)$ and was found sufficient given that the negative class is re-sampled fresh at every episode, so all classes are seen as negatives in expectation over training.

I. Optimization Procedure

Parameters θ (the full ResNet-18 stem plus both linear/conv heads) are updated with AdamW, learning rate 5×10^{-5} , weight decay 10^{-3} , under a CosineAnnealingLR schedule ($T_{\text{max}}=30$, $\eta_{\text{min}}=10^{-6}$) for up to 30 epochs with early stopping (patience 7 on validation episode accuracy). Gradients are clipped to a global norm of 5.0 before every step, which is important here because $d_c(\cdot, \cdot)$ ’s gradient magnitude grows as either argument approaches the ball boundary (the conformal factor’s singularity), so an un-clipped step taken while an embedding sits close to the boundary can eject it outside the ball; `project_to_ball` additionally re-clips every embedding produced by the exponential map after each forward pass as a second line of defense. Algorithm I-I summarizes one full training episode, corresponding to `episode_loss_and_acc` followed by an optimizer step in the reference implementation.

[t] One episodic training step of the Poincaré–Wasserstein Siamese network

Shared encoder f_θ ; curvature c ; Sinkhorn regularizer ε , iterations T_{sk} ; margin m ; loss weight λ ; N classes, K shots, Q queries/class Sample support $\mathcal{S} = \{(x_{c,k}^s, c)\}$ and query $\mathcal{Q} = \{(x_{c,i}^q, c)\}$ for the episode (Section I-A) ($g_{\mathcal{S}}, \ell_{\mathcal{S}} \leftarrow f_\theta(\mathcal{S})$; ($g_{\mathcal{Q}}, \ell_{\mathcal{Q}} \leftarrow f_\theta(\mathcal{Q})$ shared weights; global & local embeddings, Eq. 1 $c = 1, \dots, N$ $p_c \leftarrow$ Karcher mean of $g_{\mathcal{S}}$ for class c Eq. 4 $S_c \leftarrow$ position-wise mean of $\ell_{\mathcal{S}}$ for

fig1_architecture.png

Fig. 1. Dual-branch Siamese architecture. A shared ResNet-18 encoder produces a $512 \times 7 \times 7$ feature map for every support and query image. The **global branch** (red) average-pools this into one tangent vector per image, exponential-maps it onto the Poincaré ball, forms per-class hyperbolic Karcher-mean prototypes from the K support embeddings, computes the geodesic distance from each query to every prototype, and applies a softmax cross-entropy over the negated distances. The **local branch** (blue) reduces the same feature map to $P=16$ spatial points via a 1×1 convolution and 4×4 pooling, maps them onto the ball, forms per-class local prototype *sets* by position-wise averaging, and computes an entropy-regularized Sinkhorn–Wasserstein optimal-transport distance between each query’s point set and the correct/incorrect class’s prototype set, pulling toward the correct set and pushing away from a randomly sampled incorrect one by margin m . The two losses are summed with weight λ to form the total per-episode training objective.

class c each query $q \in \mathcal{Q}$ $d_{qc} \leftarrow d_c(g_q, p_c) \quad \forall c$ Eq. 2 accumulate $\mathcal{L}_{CE} += -\log \text{softmax}(-d_{q,:})_{y_q}$ sample wrong class $r \neq y_q$ uniformly $\mathcal{W}^+ \leftarrow \text{Sinkhorn}(\ell_q, S_{y_q}; \varepsilon, T_{\text{sk}})$ Eq. 6–7 $\mathcal{W}^- \leftarrow \text{Sinkhorn}(\ell_q, S_r; \varepsilon, T_{\text{sk}})$ accumulate $\mathcal{L}_{\mathcal{W}} += \mathcal{W}^+ + \max(0, m - \mathcal{W}^-)$ $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{CE} + \lambda \mathcal{L}_{\mathcal{W}}$ Eq. 9 backpropagate $\mathcal{L}_{\text{total}}$; clip $\|\nabla_{\theta}\| \leq 5.0$; AdamW step $\mathcal{L}_{\text{total}}$, episode accuracy

At evaluation time, steps 1–10 are unchanged except that support is drawn from the train-pool and queries from the held-out test-pool with augmentation disabled, no gradient is computed, and accuracy is averaged over 200 independently sampled episodes per shot value K to obtain the mean and 95% confidence interval reported in Table I; a separate full-pool pass (Section I-A) is used only to produce the confusion matrices and t-SNE embeddings, not the headline accuracy numbers.

fig1_architecture.png

Fig. 2. **Overview of the proposed Poincaré–Wasserstein Siamese few-shot framework.** A single weight-shared ResNet-18 backbone (orange, center-left) is applied identically to every image in an N -way K -shot episode — support and query alike — producing a $512 \times 7 \times 7$ convolutional feature map per image; weight sharing is what makes the network *Siamese* and guarantees that support and query embeddings are directly comparable. **Global branch (red, left):** the feature map is globally average-pooled and linearly projected to a 128-d tangent vector, then mapped onto the Poincaré ball of curvature c via the exponential map at the origin ($\exp_0^c$, Eq. 1), giving one *global hyperbolic embedding* per image. The K support embeddings of each class are combined into a *hyperbolic centroid* (Karcher / tangent-space mean, Eq. 4) to obtain per-class prototypes p_c . Each query embedding’s *geodesic* (Poincaré) distance to every prototype is computed (Eq. 2), negated, and passed through a softmax cross-entropy (Eq. 5) to obtain the classification loss $\mathcal{L}_{CE}$ — the hyperbolic analogue of Prototypical Networks. **Local branch (blue, right):** the same feature map is instead passed through a 1×1 convolution and pooled onto a coarser 4×4 spatial grid, giving $P=16$ local tangent vectors per image, which are likewise exponential-mapped onto the ball to form *local hyperbolic embeddings*. Per-class *local prototype sets* S_c are formed by position-wise averaging of the K support images’ local embeddings. Each query’s local point set is compared to a prototype set via the entropy-regularized *Sinkhorn–Wasserstein optimal-transport distance* $\mathcal{W}_\varepsilon$ (Eq. 6–7), computed with a geodesic (not Euclidean) ground cost, which finds a soft minimum-cost matching between the two 16-point sets and is therefore invariant to the spatial shift and rotation that sectioning and mounting introduce between same-subtype patches. The resulting distance is *pulled* toward the correct class’s prototype set and *pushed* away (by hinge margin m) from a randomly sampled incorrect class’s prototype set, giving the auxiliary structural loss $\mathcal{L}_{\mathcal{W}}$ (Eq. 8). **Total loss (bottom):** $\mathcal{L}_{\text{total}} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{\mathcal{W}}$ (Eq. 9) is back-propagated end-to-end through both branches into the single shared encoder every episode; no branch-specific or class-specific parameters are learned, which is what allows the trained encoder to be re-used as-is against novel support sets at test time.